

Reporting Summary

Nature Portfolio wishes to improve the reproducibility of the work that we publish. This form provides structure for consistency and transparency in reporting. For further information on Nature Portfolio policies, see our [Editorial Policies](#) and the [Editorial Policy Checklist](#).

Statistics

For all statistical analyses, confirm that the following items are present in the figure legend, table legend, main text, or Methods section.

n/a	Confirmed
☐	☒ The exact sample size (<i>n</i>) for each experimental group/condition, given as a discrete number and unit of measurement
☐	☒ A statement on whether measurements were taken from distinct samples or whether the same sample was measured repeatedly
☐	☒ The statistical test(s) used AND whether they are one- or two-sided <i>Only common tests should be described solely by name; describe more complex techniques in the Methods section.</i>
☐	☒ A description of all covariates tested
☐	☒ A description of any assumptions or corrections, such as tests of normality and adjustment for multiple comparisons
☐	☒ A full description of the statistical parameters including central tendency (e.g. means) or other basic estimates (e.g. regression coefficient) AND variation (e.g. standard deviation) or associated estimates of uncertainty (e.g. confidence intervals)
☐	☒ For null hypothesis testing, the test statistic (e.g. <i>F</i> , <i>t</i> , <i>r</i>) with confidence intervals, effect sizes, degrees of freedom and <i>P</i> value noted <i>Give P values as exact values whenever suitable.</i>
☒	☐ For Bayesian analysis, information on the choice of priors and Markov chain Monte Carlo settings
☒	☐ For hierarchical and complex designs, identification of the appropriate level for tests and full reporting of outcomes
☐	☒ Estimates of effect sizes (e.g. Cohen's <i>d</i> , Pearson's <i>r</i>), indicating how they were calculated

Our web collection on [statistics for biologists](#) contains articles on many of the points above.

Software and code

Policy information about [availability of computer code](#)

Data collection	Behavioral data collection was conducted through the behavioral experiment described in the manuscript. The experiment was implemented using custom-made code, which is available on the competition's website.
Data analysis	Data analysis was performed using MATLAB 2023a. The code is publicly available: https://doi.org/10.5281/zenodo.14548839

For manuscripts utilizing custom algorithms or software that are central to the research but not yet described in published literature, software must be made available to editors and reviewers. We strongly encourage code deposition in a community repository (e.g. GitHub). See the Nature Portfolio [guidelines for submitting code & software](#) for further information.

Data

Policy information about [availability of data](#)

All manuscripts must include a [data availability statement](#). This statement should provide the following information, where applicable:

- Accession codes, unique identifiers, or web links for publicly available datasets
- A description of any restrictions on data availability
- For clinical datasets or third party data, please ensure that the statement adheres to our [policy](#)

Data files can be found on the competition website: <https://sites.google.com/view/cec19/home> as well as the competition's repository: <https://doi.org/10.5281/zenodo.14548839>

Research involving human participants, their data, or biological material

Policy information about studies with [human participants or human data](#). See also policy information about [sex, gender \(identity/presentation\), and sexual orientation](#) and [race, ethnicity and racism](#).

Reporting on sex and gender	No demographical information was collected.
Reporting on race, ethnicity, or other socially relevant groupings	No demographical information was collected.
Population characteristics	No demographical information was collected.
Recruitment	Data was collected via the online labor platform Amazon Mechanical Turk. Exclusion criteria were based on task performance.
Ethics oversight	The study was approved by the Ethics Committee of the Psychology Department at the Hebrew University of Jerusalem, Israel.

Note that full information on the approval of the study protocol must also be provided in the manuscript.

Field-specific reporting

Please select the one below that is the best fit for your research. If you are not sure, read the appropriate sections before making your selection.

☐ Life sciences ☒ Behavioural & social sciences ☐ Ecological, evolutionary & environmental sciences

For a reference copy of the document with all sections, see nature.com/documents/nr-reporting-summary-flat.pdf

Behavioural & social sciences study design

All studies must disclose on these points even when the disclosure is negative.

Study description	Participants were presented with two choice alternatives, displayed as rectangles randomly positioned on either the left or right side of the screen and randomly colored red or blue. They were instructed to make sequential choices between these alternatives and were informed that some choices could yield rewards. The number of remaining trials was displayed on the screen, starting at 100 and decreasing by one after each choice. Rewards were represented by smiley face icons that accumulated in a dedicated section of the screen, indicating the total rewards earned. Choices that did not result in rewards were marked by a sad smiley icon, which did not accumulate. The experiment ended after 100 trials.
Research sample	No demographic information was collected from the participants.
Sampling strategy	To identify the most effective schedule, we utilized the method of Successive Rejects (Audibert and Bubeck, 2010). According to this method, more participants (more "samples") are allocated to better-performing reward schedules. The method of Successive Rejects operates iteratively. In the first iteration, all reward schedules are allocated an equal number of participants. Then, the worst performing schedule (the schedule associated with the minimal bias) is "rejected" in the sense that no more participants are allocated to it. The process continues iteratively until just two schedules remain. The schedule that yields the highest bias over all tested schedules is considered the most effective one. According to this process, the number of participants that should be used in each iteration depends on (1) the total number of schedules tested, (2) the (unknown) distribution of their "true" biases and (3) the desired confidence level for identifying the most effective one. We pre-determined the number of participants per iteration assuming the true bias of the schedules is evenly distributed between 50% (chance level) and 70%. One minor complication when running online experiments is that not all the participants who start the task also end up completing it. We randomly allocated participants to schedules in advance, which resulted in some variability in the number of participants that completed the task per schedule. To make sure that each schedule is tested on no-fewer than the number of participants required by the Successive Reject method, based on pilot data, we pre-allocated 30% more participants than required to each schedule.
Data collection	Data was collected through an online experiment that participants could access via a web browser.
Timing	Data collection began on May 25, 2020, and concluded on February 23, 2021.
Data exclusions	Participants who selected either alternative fewer than five times were excluded from the analysis.
Non-participation	The data of participants who did not complete the experiment was not recorded.
Randomization	According to the "successive rejects" method, the competition was conducted in several iterations, each including a specific set of reward schedules. In each iteration, participants were randomly assigned to the current set of reward schedules.

Reporting for specific materials, systems and methods

We require information from authors about some types of materials, experimental systems and methods used in many studies. Here, indicate whether each material, system or method listed is relevant to your study. If you are not sure if a list item applies to your research, read the appropriate section before selecting a response.

Materials & experimental systems

n/a	Involved in the study
☒	☐ Antibodies
☒	☐ Eukaryotic cell lines
☒	☐ Palaeontology and archaeology
☒	☐ Animals and other organisms
☒	☐ Clinical data
☒	☐ Dual use research of concern
☒	☐ Plants

Methods

n/a	Involved in the study
☒	☐ ChIP-seq
☒	☐ Flow cytometry
☒	☐ MRI-based neuroimaging

Plants

Seed stocks	NA
Novel plant genotypes	NA
Authentication	NA